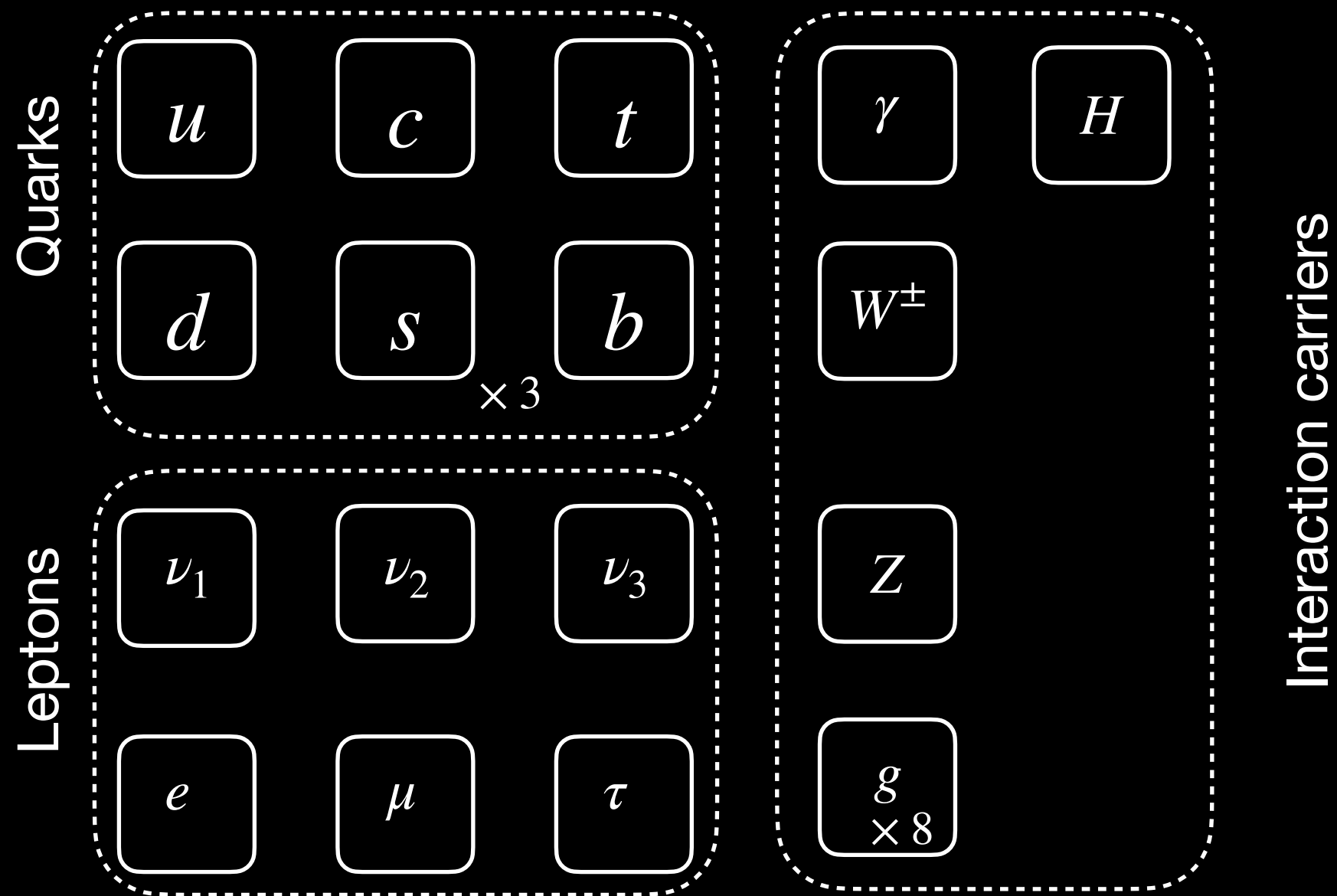


Standard Model (updated)



Mixing of Lepton Generations

- ★ Interaction amplitude proportional to $\frac{g}{2\sqrt{2}}V_{\alpha i}^*$ for ν_i and ℓ_α
- ★ Nine numbers $V_{\alpha i}^*$ make unitary lepton mixing matrix PMNS

$$V = \begin{pmatrix} V_{e1}^* & V_{e2}^* & V_{e3}^* \\ V_{\mu 1}^* & V_{\mu 2}^* & V_{\mu 3}^* \\ V_{\tau 1}^* & V_{\tau 2}^* & V_{\tau 3}^* \end{pmatrix}$$

Pontecorvo-
Maki-
Nakagawa-
Sakata

- ★ Nondiagonal form of the mixing matrix lead to neutrino oscillation